

Code Boilerplate generator (cbp-gen)

Project description

A native application that allows me to create boilerplate code for any programming language, C/C++, Go, React, and more in the future.

Requirements

- Must be able to run on all computer operating systems (MacOS, Unix/Linux, Windows)
- Should have a front-end that is capable of connecting with a back-end
- Should use a modern, fast language to quickly generate the boilerplate
- Must allow user inputs to customize the boilerplate code
- For minimum viable product, must be able to create boilerplate for these languages:
 - Go, C, C++, HTML, React, JavaScript
- Can run application from anywhere on the system and be able to create boilerplate code in specific correct locations
- Can access the terminal/command line to be able to run commands such as “npm install”.
- Allow for user customizability with what they want in their boilerplate code
- This application should be able to scale with multiple new languages for boiler plates

What language to use:

Tauri (React front-end + Rust back-end):

- Pros:
 - Looks great on a resume
 - Learn a new programming language
 - Close to industry standard
 - React front-end capabilities
 - Works on all platforms
- Cons:
 - Probably a lot of AI learning which sucks
 - I am not familiar with Rust

C#/.NET + Avalonia:

- Pros:
 - Works on all platforms
 - Learn .NET architecture (good for resume)
 - Add C# to resume
 - New front-end tech

- Good for industry
- Cons:
 - Need to learn a new language
 - C# code is hard to write and feels super bloaty

Weils (React front-end + Go back-end):

- Pros:
 - I want to learn Weils
 - I already know Go so no new learning
 - Weils uses js front-end frameworks
 - Runs on all platforms
- Cons:
 - Go is more niche
 - No new industry tech to add to resume

Electron (React front-end + Node back-end)

- Pros:
 - Already familiar with JavaScript
 - React front-end
 - Simple to start
- Cons:
 - Takes up a lot of memory due to Chromium being packaged
 - Do not want to work with Electron

Agile Sprints

First Sprint:

- ~~Pick a programming language~~
- ~~Make a new file in the current directory~~
- ~~Make a file using the commandline/shell~~
- ~~Run a commandline/shell command~~
- ~~Detect if a file already exists~~
- ~~Detect which OS is benign run on the system~~
- ~~Learn how file paths work using Go and changing file paths~~

Second Sprint:

- Implement a template format for C
- Write a basic CLI styled program for the boilerplate generator
- Learn how to make flags while using Go
- Allow customization for user:
 - What type
 - Change the name

Third Sprint:

- Make the front-end
- Implement a feature to allow user inputs
- Add a new template for another language (probably Go)

Fourth Sprint:

- Implement a React boilerplate that allows for multiple options. We want to use the npm installer for now

Fifth Sprint:

- Update the front-end to be nicer.
- Make a website to host the project for people to be able to download